Dynamic Learning and Optimal Advertising Mechanism for LLM Platforms

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Reading Assignment

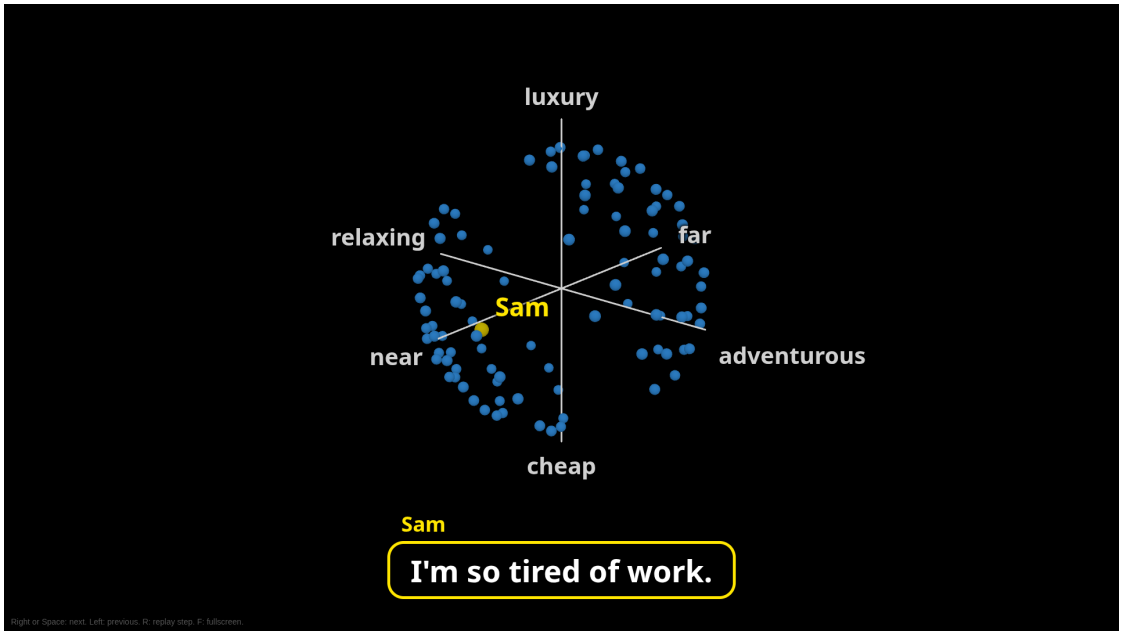
CS1.408 Introduction to Game Theory

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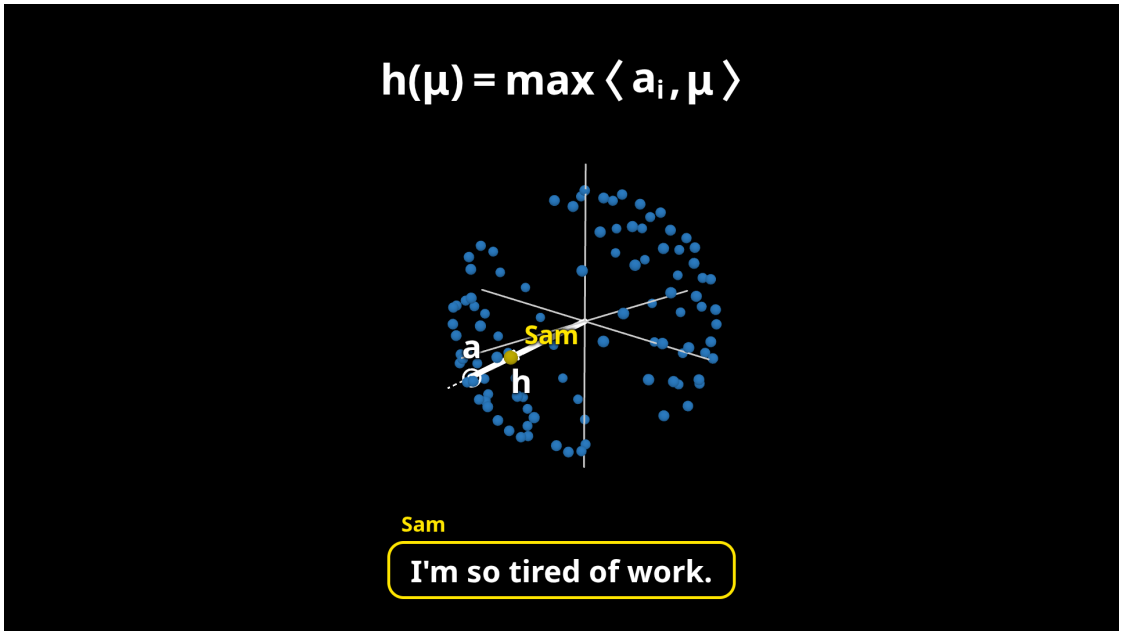
The problem: one ad slot



Sam, the ads and the bot in aspect space

- Sam chats with a bot
- The bot can show **one ad**
- Ads and Sam are points in aspect space

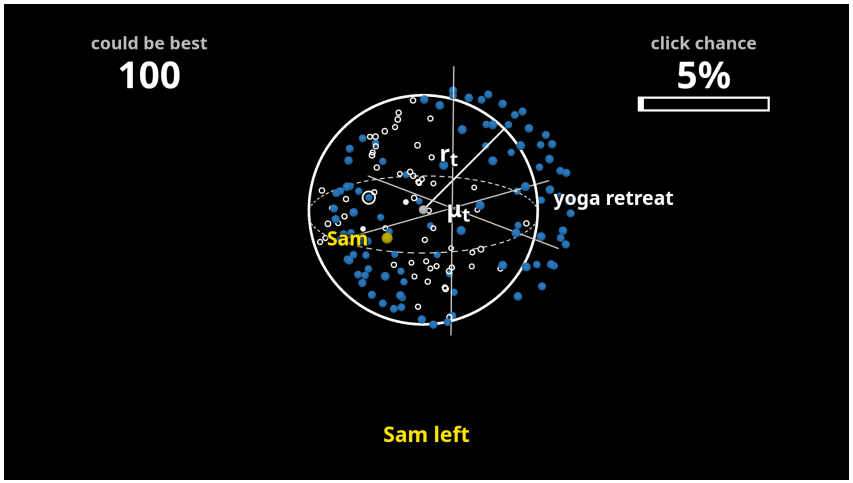
Fit and bids



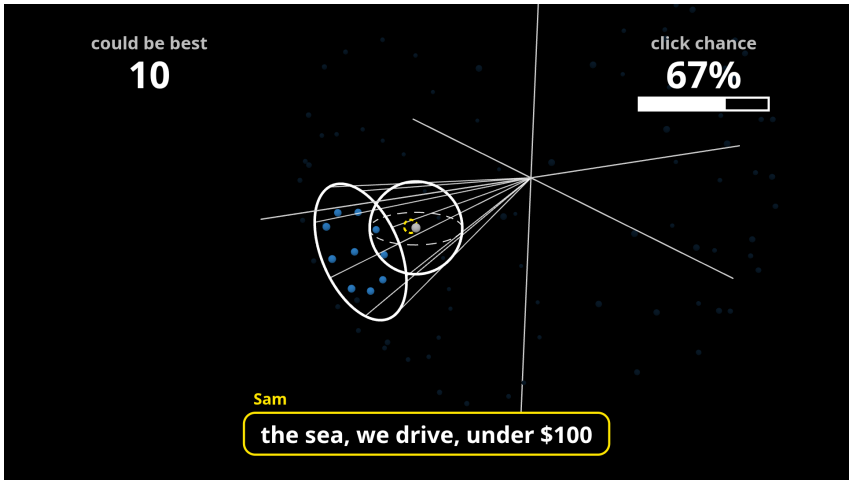
Fit of an ad is its dot product with Sam

- Fit $= \langle a, \mu \rangle$
- Best fit $h(\mu)$: top ad, or 0 for no ad
- A bid v pushes an ad further out

Show now or wait?



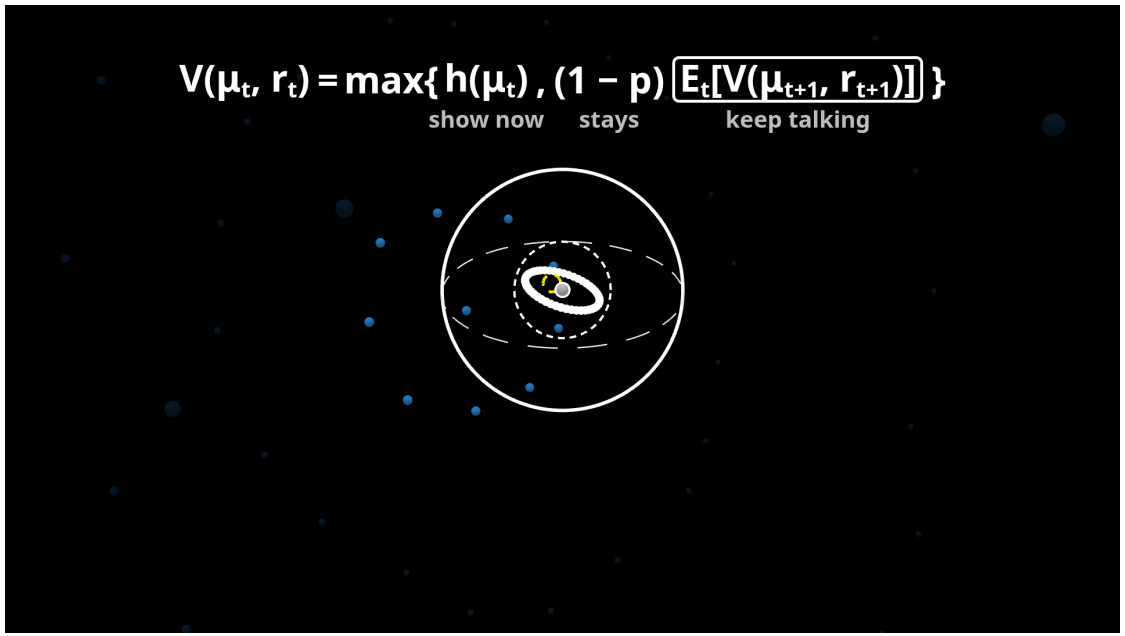
Show now: click chance 5 in 100



Keep talking: ads 100, 50, 30, 10, click chance 2 in 3

The bot only has a guess μ_t and a ball of radius r_t. Each message shrinks it by α.

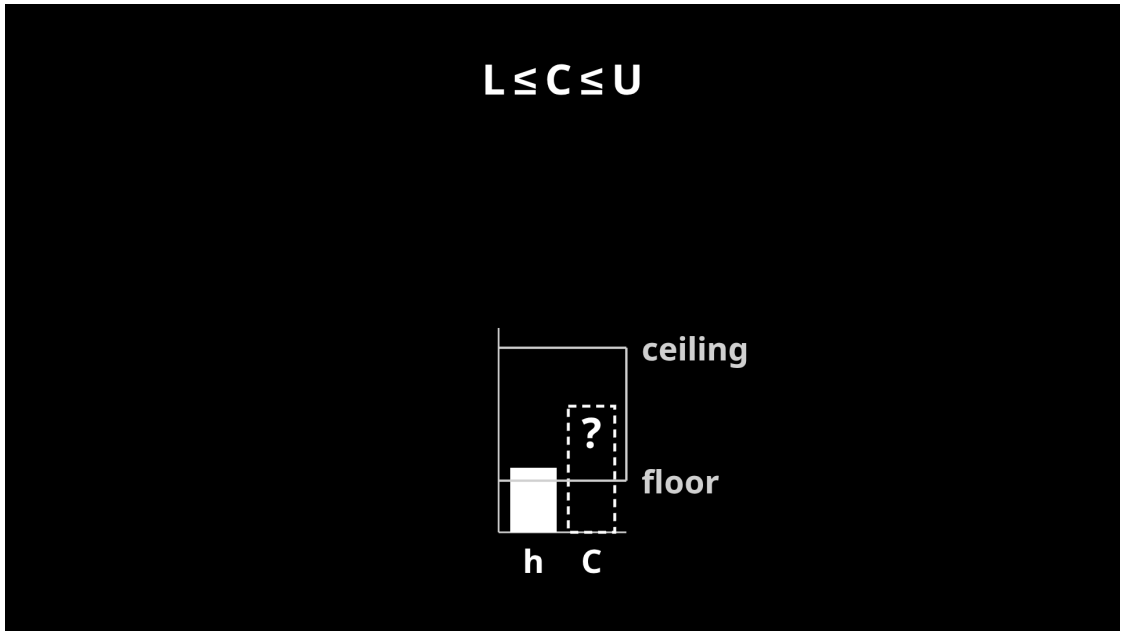
The catch



Best plan $V = \max(\text{show now}, \text{wait})$

- Sam may leave after any message, chance $p = 0.2$
- Waiting needs the spread of the next guess
- The bot never has it

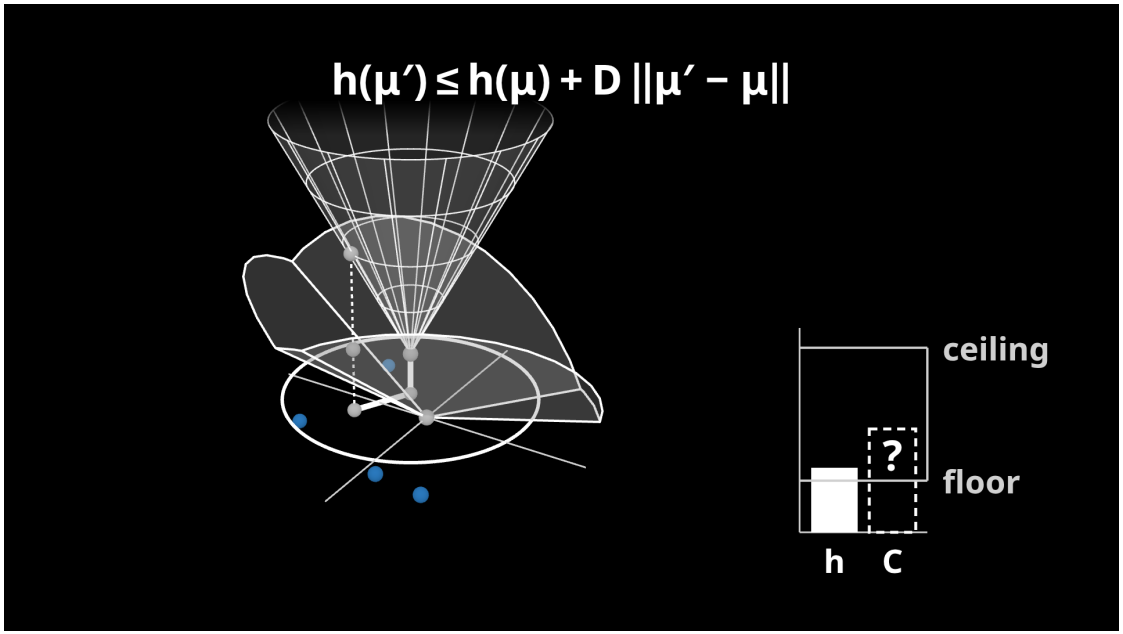
Fence the worth of waiting



Meter: h is show now, C is the hidden worth of waiting

- Exact planning breaks
- Use a floor L and a ceiling U instead

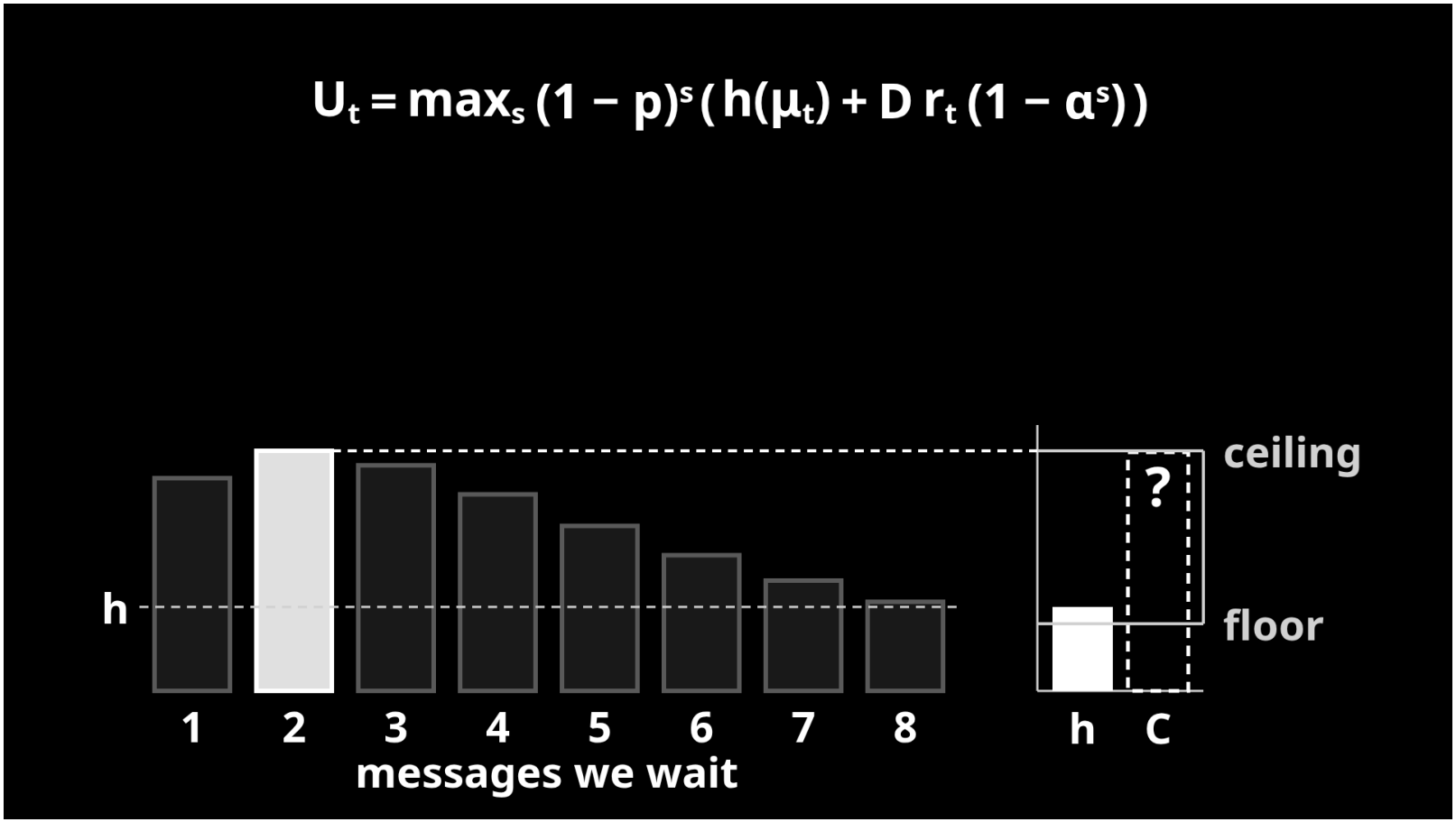
Floor and ceiling



Gap D between pushed ads, bowl slope at most D

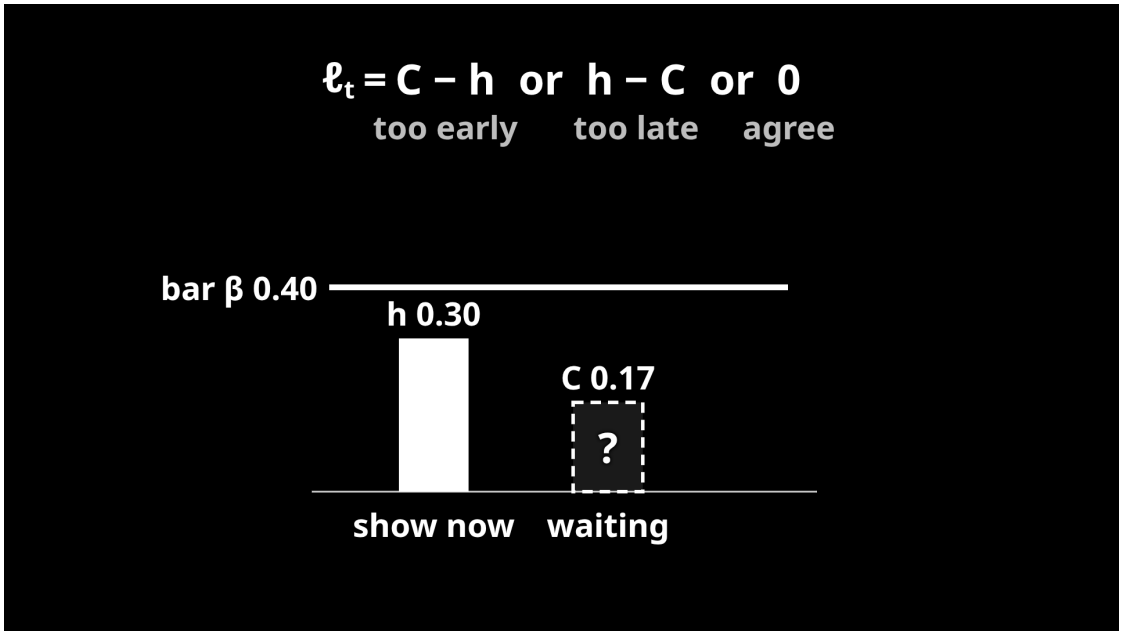
- Floor $L = (1 - p)\, h$
- D: longest gap between pushed ads
- The cone covers the bowl of fit

The ceiling



One bar per wait length, shrunk by $(1-p)^s$. The tallest is the ceiling $U = C\,D\,r$.

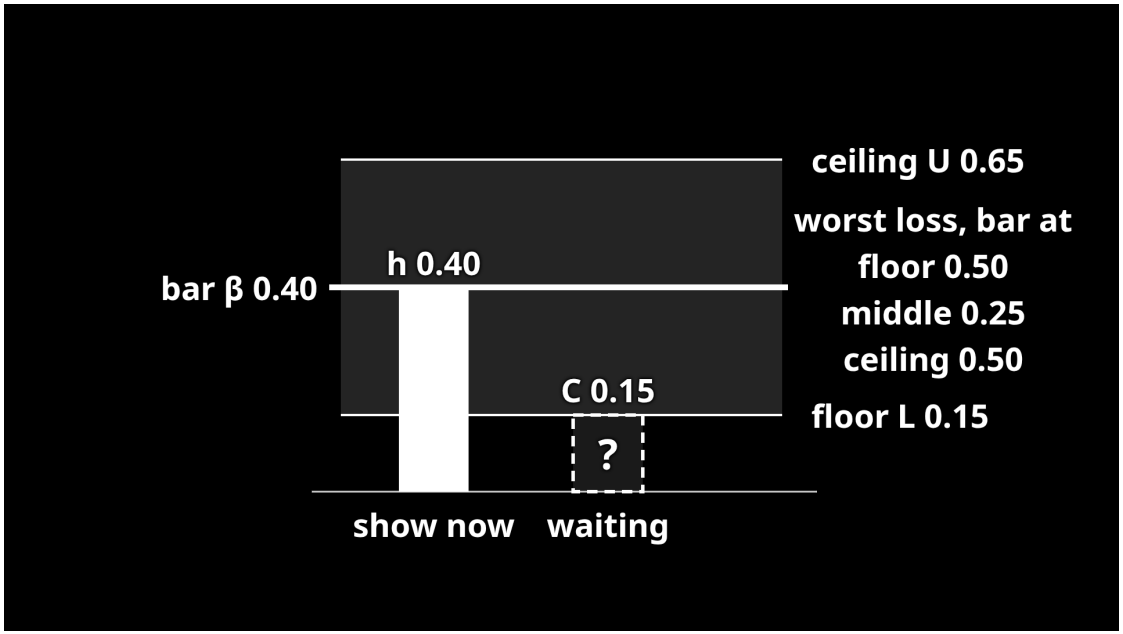
Stop at the middle



Bar too low shows early. Bar too high waits too long.

- Bar below h: talk
- Bar reaches h: show
- Wrong bar costs loss

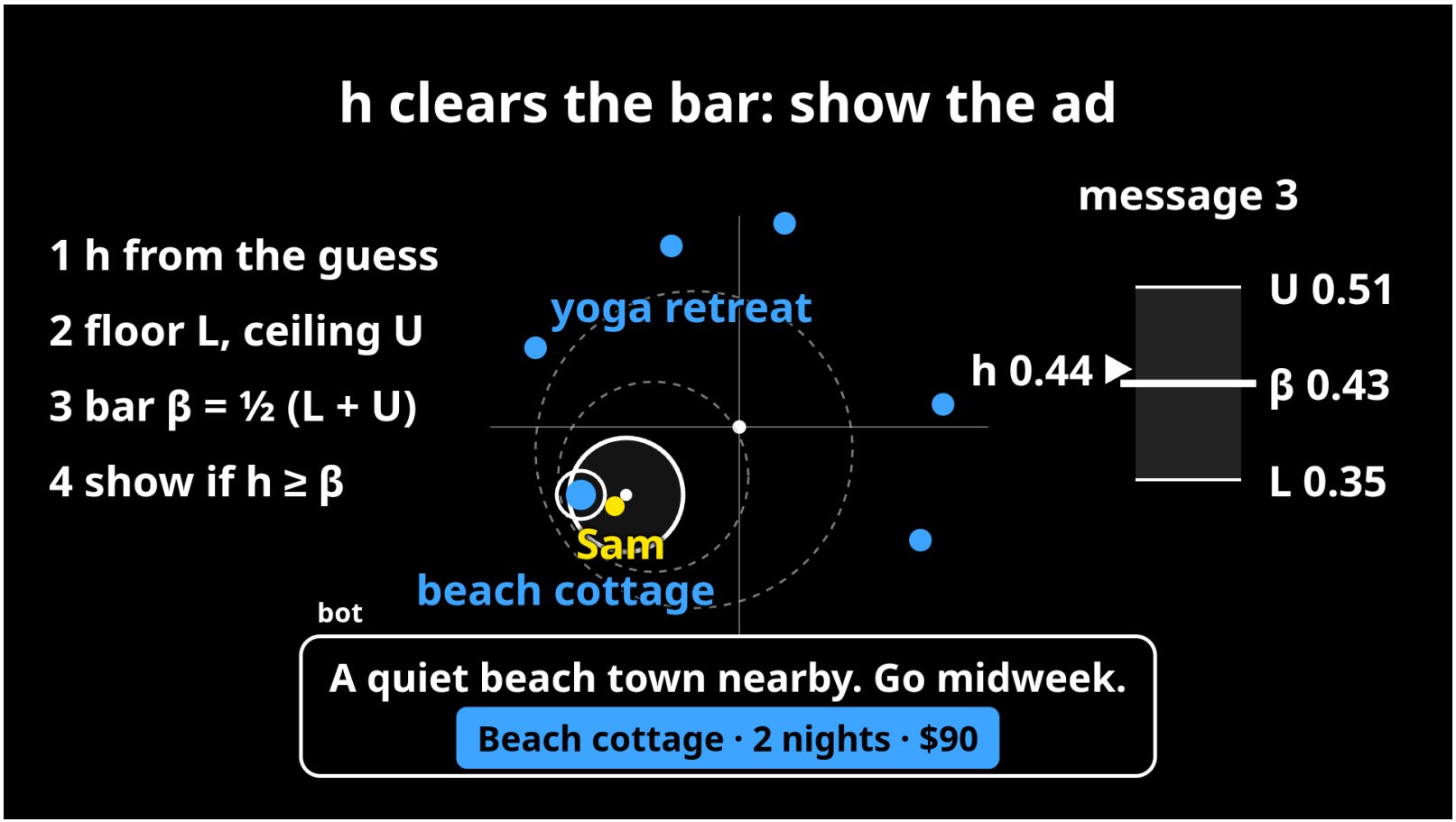
Why the middle



Worst loss drops from 0.50 to 0.25

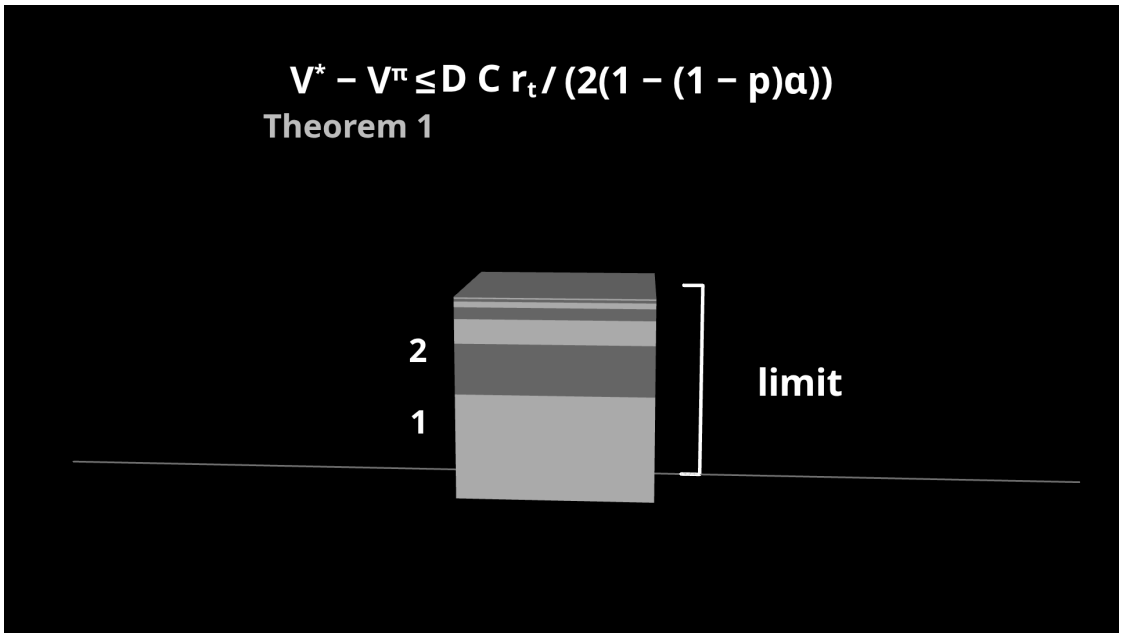
- Bar at the ceiling: wrong by the whole fence
- Bar at the midpoint: wrong by half
- Algorithm 1 uses the midpoint

Sam's chat



Message 3: $h = 0.44$ reaches the bar 0.43, so show the beach cottage.

Theorem 1



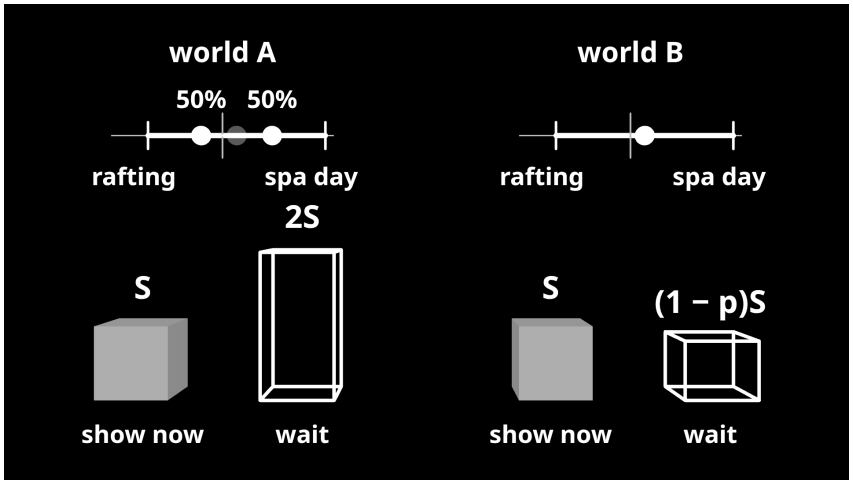
Each wait is worth $(1-p)\alpha$ of the last

Welfare loss

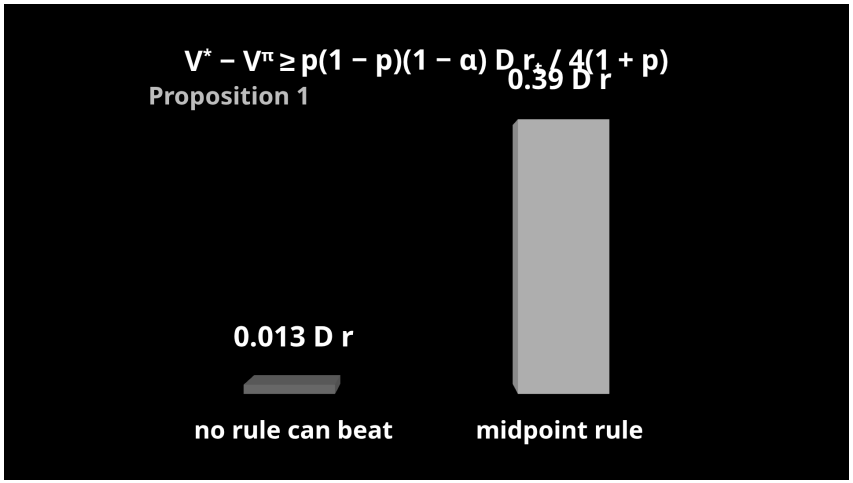
$$\leq \frac{D\,C\,r}{2\,(1-(1-p)\alpha)}$$

- Sam: about $0.39\,D\,r$

No rule does much better



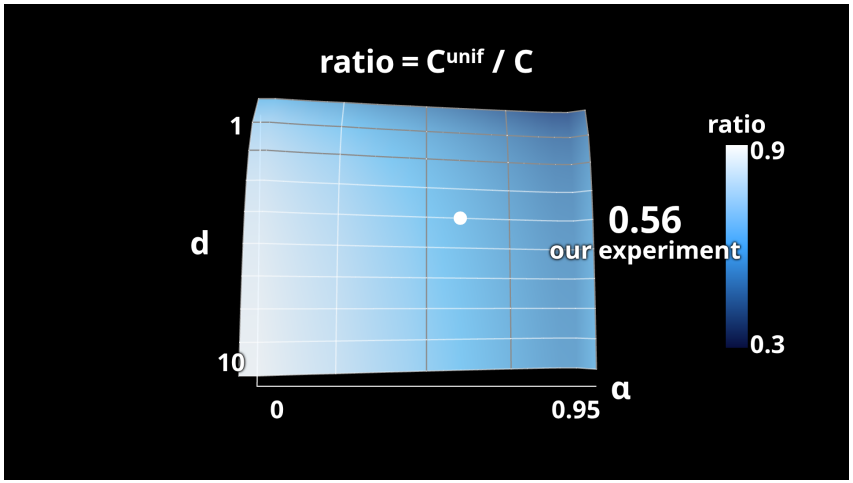
Two worlds look the same today



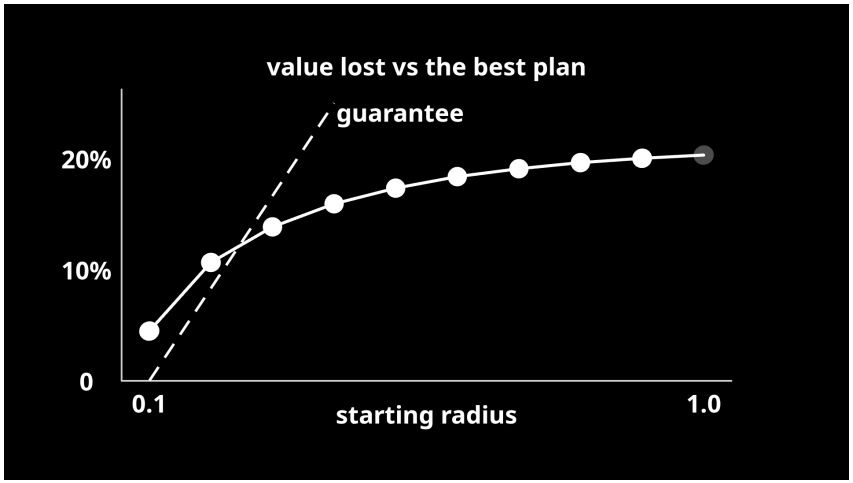
Proposition 1: no rule beats $0.013\,D\,r$

Midpoint gets $0.39\,D\,r$. Same shape, same $C\,D\,r$.

Simulations



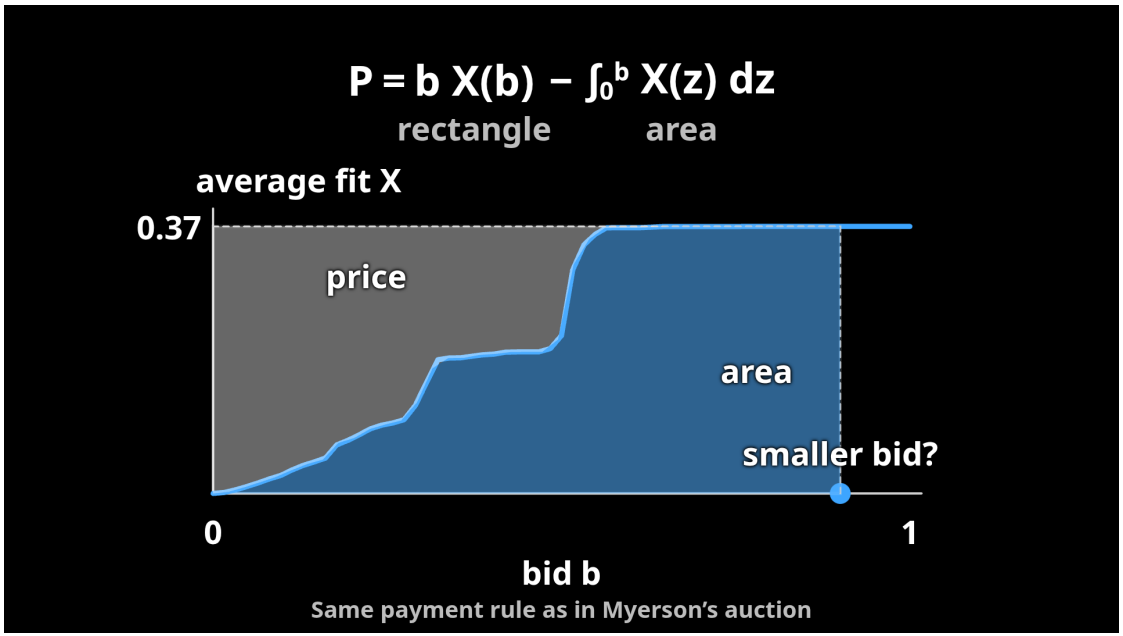
Even spread ceiling C_{unif}, ratio 0.29 to 0.91



Random markets, radius 0.4

The policy keeps 84 percent of welfare, far under the guarantee.

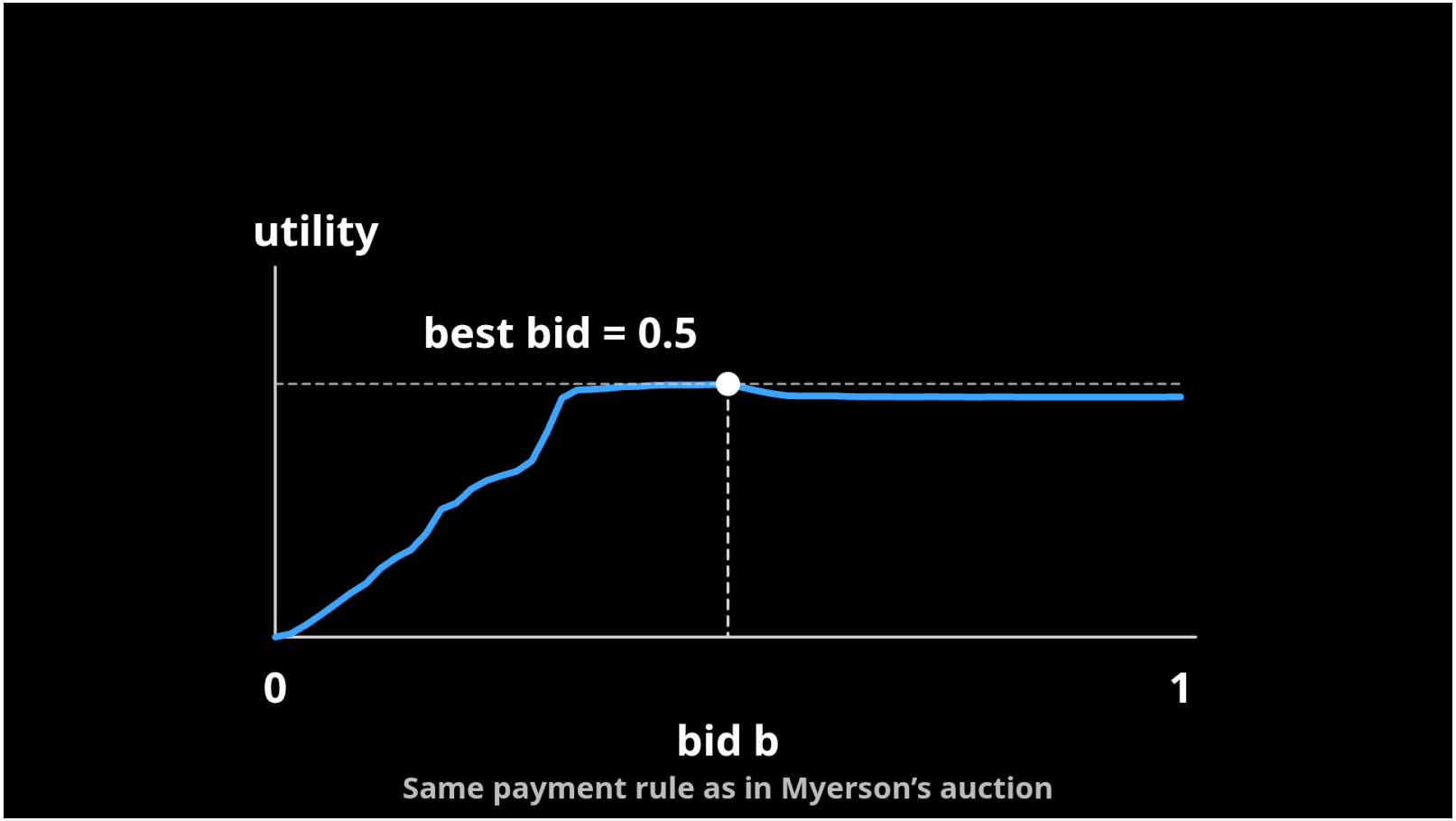
Honest bids



Pay $= b\,X(b)$ minus the area under X

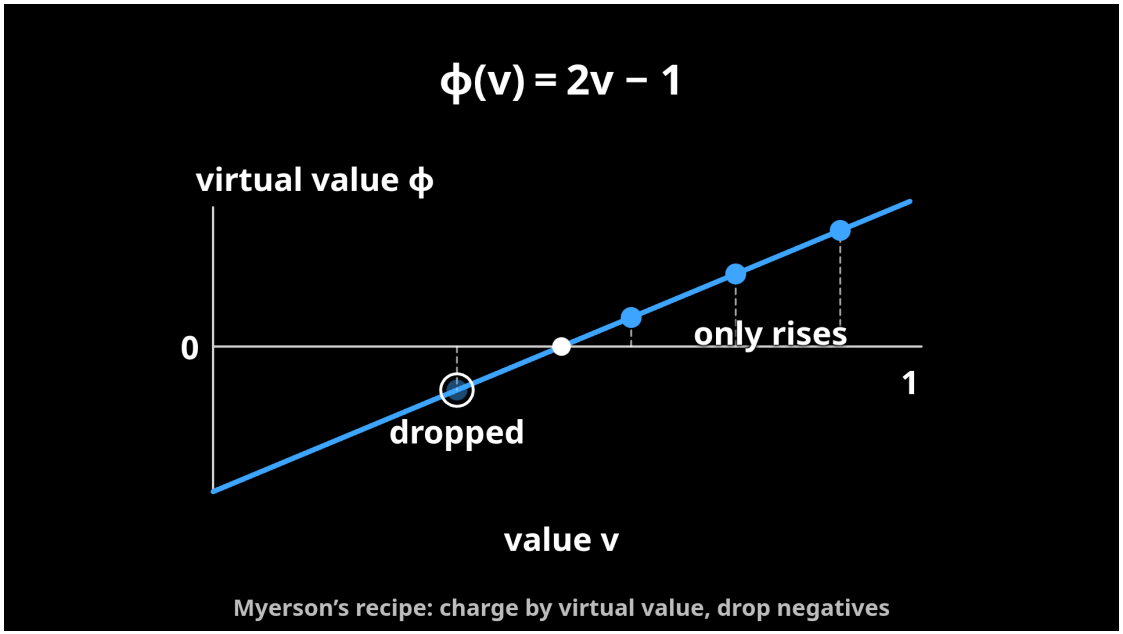
- Bid once before the chat
- $X(b)$: average fit at bid b, only rises
- Payment is the area under the staircase

Bidding true value is best



Low or high bids lose a sliver. Utility peaks at the true value, 0.5.

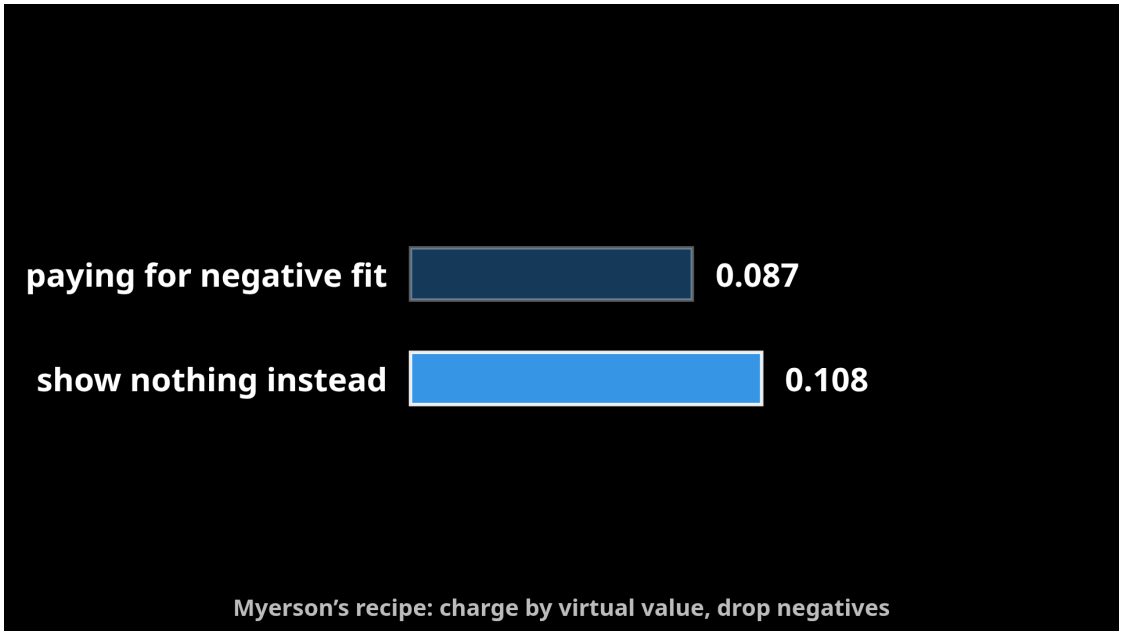
Virtual value



Rafting, value 0.35, scores -0.3 and is dropped

- $\varphi(v) = v - \frac{1-F(v)}{f(v)}$
- Even values: $2v - 1$
- Same recipe as Myerson's auction (the label is ours, the paper cites Myerson 1981 only in related work)

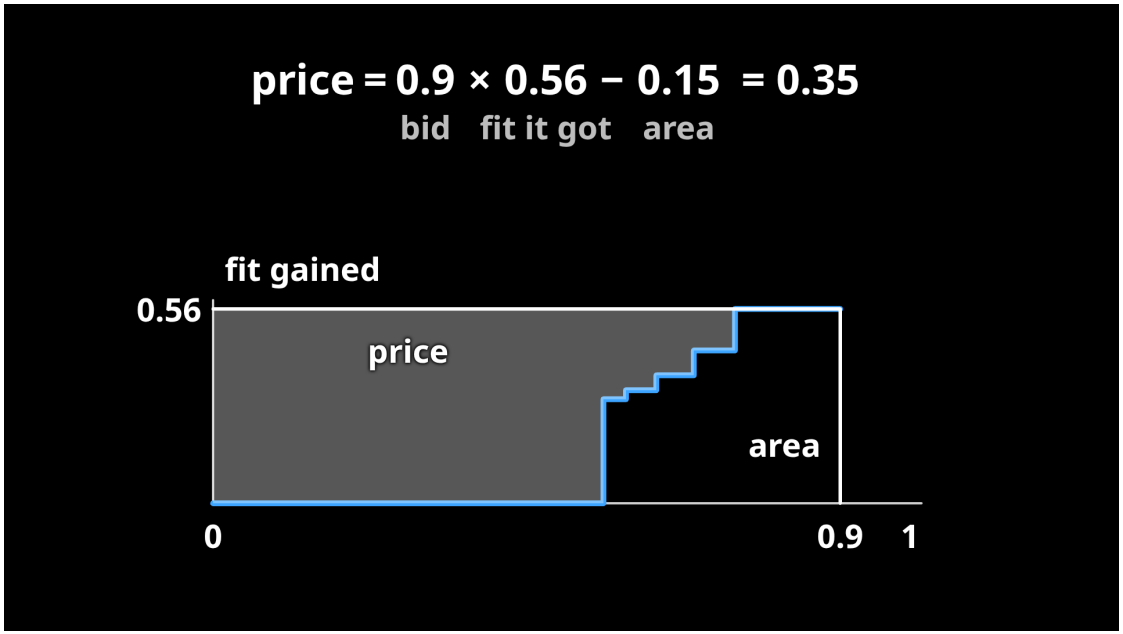
Lemma 4 and Algorithm 2



Average pay equals average φX, both 0.0776

1. Drop ads with $\varphi < 0$
2. Push the rest by φ
3. Run the same chat rule

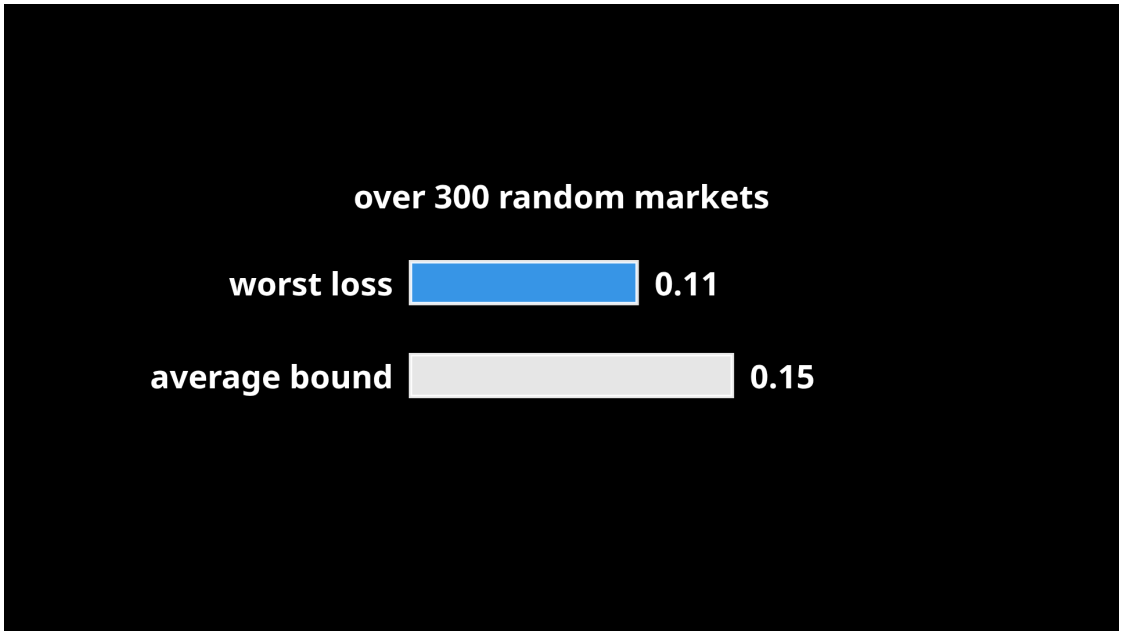
Price from one chat



Staircase of 5 steps from one replayed chat

- The platform sees one chat only
- Replay it with lower pushes
- Price $= 0.9 \times 0.56 - 0.15 = 0.35$

Theorem 2



300 random markets: worst loss 0.11, bound 0.15 on average

Revenue loss

$\leq D_+^{\varphi}\, C\, r_1$

- Honest and individually rational
- D_+^{φ}: virtual spread

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Takeaway and next steps

Takeaway

- Bid, push by φ, talk while unsure, charge the area
- Computable and honest
- Near best revenue

Next

- Budgets
- Real chats
- Ads that change Sam
- Length limits

References I

S. Alaei, A. Makhdoumi, and A. Malekian. Dynamic learning and optimal advertising mechanism for LLM platforms. 2025.

R. B. Myerson. Optimal auction design. *Mathematics of Operations Research*, 6(1):58–73, 1981.